UNIVERSITI TEKNOLOGI MALAYSIA

LAB REFLECTION #2 Team 1 (7.5%)

SEMESTER II 2021/2022

<<LIST ALL TEAM MEMBERS' NAME WHO HAVE PARTICIPATED IN THE LAB ONLY>>

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CODE OF SUBJECT: SECJ3403

NAME OF SUBJECT: SPECIAL TOPIC IN SOFTWARE ENGINEERING

OBJECTIVES:

1. Identify the important concepts in cloud computing.
2. Do reflection on practical lab implementation based on the concepts identified.

INSTRUCTION:

1. Create reflection on practical lab implementations for the following labs:
    i. Database Lab 5 - Build a Database Server
    ii. Automatic Scaling and Monitoring Lab 6 - Scale & Load Balance your Architecture

Reflection on practical lab implementations (based on Gibbs' Reflective Model)

To do reflections on practical lab implementation, the following templates should be used. You have to use the same templates for all lab reflections.

Reflection on Database Lab 5 and Automatic Scaling and Monitoring Lab 6	
Description	Describe in brief what you did in the lab in your own words. ● In this lab, we were introduced to accessing the AWS management console to create a security group for the RDS DB instance. ● A step-by-step guide is provided to explain in more detail what this lab tries to accomplish. ● We followed the instructions in lab 5 to build a database server module and found it to be easy to follow. ● A successful lab-db database is created at the end of the lab session.
Evaluation	Describe the positive or negative experience during the lab ● Hands-on experience by individuals is importantly gained by the student itself. ● A good introduction starts for a student that does not have any experience in managing cloud architecture. ● A complete guide helped students to be at ease in completing the lab. ● The positive experience of trying the lab was discovering features provided by AWS that I can replicate for my own application. The other positive experience was learning how to use the console and familiarizing myself with AWS UI. ● The negative experience was searching for the services menu that was described in the instruction. Since we are new to AWS UI, we need to press ctrl+f and type service to find it. Another negative experience was the waiting time for the database deployment. It feels like more than 4 minutes.

	• The load balancing instruction was confusing and we had to redo the step to keep up with the instruction.
Analysis	ii. Which practical implementation in the lab can be related to the theory in the lecture? <u>Database</u> • The lecturer did explain everything in detail about all the things that are available in the console for example VPC (virtual private cloud), database type, subnet group, RDS (relational database service), and DB instances • The lab explained how we can store, maintain, and access any type of data similar to what the lecturer taught. • All contents in the learning lab related to the theory given during lecture discussion and helped us to set up a database in AWS. <u>Load Balancing</u> • The lecturer explains everything we need to know about the process of distributing network traffic across multiple AZ. • The lab explained how load balancing improves application and prevents server overload similar to what the lecturer taught. • All contents in the learning lab related to the theory given during lecture-discussion and helped us to set up Load Balancing in Amazon ii. What other concepts in your opinion should be included in the lab implementation? <u>Database</u> • A NoSQL database concept should be included because we can learn how to work with a NoSQL database in AWS. • Need to show other database types and their implementation and which scenario it is supposed to be used.

	• Need more detailed explanation on how to properly create the database so we can understand easily <u>Load Balancing</u> • I don't know much about networking and load balancing so I am satisfied with the features provided. • Tutorial labs of different load balancing are quite important to be shown so it can help us as students to fully understand and be able to decide on which scenario or situation that specific load balancing meets the criteria to be chosen. • Need a more detailed explanation of how load balancing works
Conclusion	Focus on what you have learned. Are there any skills you developed because of the experience? If so, how would you apply them in future experiences or situations? • By learning in this lab, students are able to develop new knowledge in Amazon RDS not just in theory but also in practical use. Help to boost productivity and creativity in the current work environment in the IT field. • From this experience, students are able to deploy Amazon RDS which can be an option for a company. • We are able to familiarize ourselves with the AWS services. If we ever have the chance to use cloud services, we might as well use AWS because we can leverage our experience from this course to help accelerate our future projects. • Students can now understand how to set up, operate, and scale databases in the cloud with Amazon RDS
Action Plan	What specific implementation in this lab should be included in your proposed team project? Project: Mersing tourism

	Database ● We might use a NoSQL database that is cloud-based for our project. It will be preferred if the cloud service provided a free version that we can use. ● At the moment, there is no implementation in AWS microservice for the work environment. Lab practices are needed to maintain the skill developed during this lab. If we are able to extend the use of this AWS Cloud Foundation, we may be able to fully understand the AWS ecosystem and help us move our project to AWS. Load Balancing ● We will not use a load balancing method for our project since we will focus only on using a single server. ● Our current situation is we use monolithic architecture where we store all source code in a single same server as our web server. If we are able to prolong lab practice to fully understand the AWS ecosystem and if our current single server gains more traffic than usual, we may consider using load balancing in the future.

Rubric for reflection

Criteria	Excellent	Good	Fair	Unsatisfied
Evaluation **(5 marks)**	Clearly explains positive experiences during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work. **(4-5 marks)**	Somewhat explains positive experiences during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work. **(3-3.9 marks)**	Little explanation on positive experiences during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work. **(2-2.9 marks)**	No clear explanations of positive experience during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work. **(0-1.9 marks)**
Analysis **(10 marks)**	Analysis shows a strong understanding between the concept and the implementation **(8-10 marks)**	Analysis shows mostly strong understanding between the concept and the implementation. **(5 -7.9 marks)**	Analysis shows little evidence of understanding between the concept and the implementation **(3 -4.9 marks)**	Analysis shows no evidence of understanding between the concept and the implementation **(0 -2.9 marks)**
Conclusion **(5 marks)**	Conclusion shows strong evidence of synthesis of ideas presented and insights gained	Conclusion mostly shows strong evidence of synthesis of ideas presented and insights gained	Conclusion shows little evidence of synthesis of ideas presented and insights gained throughout the entire course. **(2-2.9 marks)**	Conclusion shows no evidence of synthesis of ideas presented and insights gained throughout the entire course.

	throughout the entire course. **(4-5 marks)**	throughout the entire course. **(3-3.9 marks)**		**(0-1.9 marks)**
Action Plan **(5 marks)**	Thoroughly explains the importance of the action and future implementation in the group project. **(4-5 marks)**	Somewhat explains the importance of the action and future implementation in the group project. **(3-3.9 marks)**	Unclearly explains the importance of the action and future implementation in the group project. **(2-2.9 marks)**	No explanation of the importance of the action and future implementation in the group project. **(0-1.9 marks)**
Writing of the Reflection **(5 marks)**	Writing is clear, concise, and well organized with excellent sentence/paragraph construction. Thoughts are expressed in a coherent and logical manner. There are no more than three spelling, grammar, or syntax errors per page of writing. The documentation is in a systematic and formatted manner.	Writing is mostly clear, concise, and well organized with good sentence/paragraph construction. Thoughts are expressed in a coherent and logical manner. There are no more than five spelling, grammar, or syntax errors per page of writing. The documentation is in a somewhat systematic and formatted manner. **(3-3.9 marks)**	Writing is unclear and/or disorganized. Thoughts are not expressed in a logical manner. There are more than five spelling, grammar, or syntax errors per page of writing. The documentation is a little systematic and formatted manner. **(2-2.9 marks)**	Writing is unclear and disorganized. Thought's ramble and make little sense. There are numerous spelling, grammar, or syntax errors throughout the response. The documentation has no systematic and formatted manner. **(0-1.9 marks)**

	(4-5 marks)			